Internal study workflow and evaluated model variants

Input Datasets

Omics Datasets

- Gene Expression
- DNA Methylation

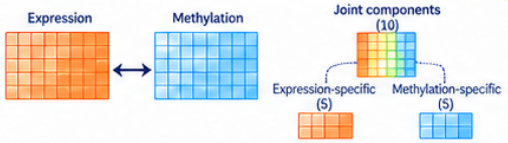
Chemical Structure

- Drug Fingerprints (binary)

Latent representations

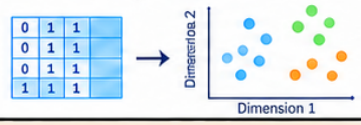
O2PLS (main framework)

10 joint scores (expression side)
5 expression-specific + 5 methylation-specific



MCA: Multiple correspondence analysis

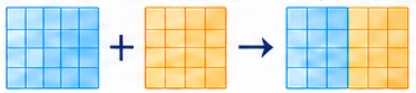
20 drug coordinates



Integration

Feature concatenation

One omics representation + drug coordinates
Main comparisons: 20 + 20 = 40 features
Scaling fitted on training data



Additional variants

- O2PLS component ablation
- Tissue covariate in selected PCA branches
- Feature count varies by branch

Modelling

XGBoost Regressor

Prediction

ln(IC50 / 1 µM)

Evaluation

Held-out cells; known drugs

Training target

Observed ln(IC50 / 1 µM)

Supervised training only

Validation procedure (internal study)

Cell-level split → Five-fold inner CV → Freeze selected configurations → Held-out evaluation

48 XGBoost configurations per tuned branch; selection by pooled OOF RMSE.

Baselines (separate from XGBoost routes)

- drug mean
- shrunken drug–tissue mean

Study-specific scaling and representation fitting use training data only.

The published omics panel was prefiltered before this study.